

Options Greeks Dashboard

A Black-Scholes-based options Greeks dashboard for NIFTY / BANKNIFTY, built as an S&T interview portfolio project. Live NSE option chain data with a deterministic sample-data fallback for offline/demo use.

Features

- **Chain View** — Delta, Gamma, Theta, Vega for calls and puts across the full strike chain, plus a plot of any Greek vs. strike (smile-shaped curves).
- **Single Option** — Deep-dive on one strike/type: theoretical price, all five Greeks, IV backed out from the live LTP via Newton-Raphson, and a payoff diagram.
- **Portfolio Builder** — Add multiple legs (e.g. Buy Call + Buy Put = long straddle) and see aggregated **net Delta/Gamma/Theta/Vega/Rho** — this is the same shape of risk view used on a live trading desk — plus a combined P&L-at-expiry chart.

Setup

```
bash

pip install -r requirements.txt
streamlit run app.py
```

Data

- Attempts a live pull from NSE's option chain API via `nsepython`.
- NSE aggressively rate-limits/blocks non-browser and many cloud/hosted IPs, so if the live fetch fails, the app **falls back to a sample chain** automatically (clearly labeled in the sidebar:  live vs  sample) so the dashboard is always demoable.
- Run locally (laptop/college wifi) for the best shot at live data; a hosted demo will likely show sample data — that's expected and disclosed in-app.

Methodology notes (for interviews)

- Pricing: standard Black-Scholes-Merton, no early exercise (reasonable for European-style NSE index options).
- Greeks are reported in trader-convention units: Theta per calendar day, Vega per 1 vol point, Rho per 1% rate move.
- IV solved via Newton-Raphson with a bisection fallback for cases where Newton fails to converge (deep ITM/OTM, near-expiry).

- Risk-free rate is a manual input (default ~6.5%, proxying the short-end India T-bill rate) — flagged as a simplification; a real desk would use a proper OIS/MIBOR curve.
- No dividend yield adjustment needed for index options ($q=0$ by default, though the pricing functions support q for extensibility to stock options).

Possible extensions

- Pull a real risk-free curve instead of a flat manual rate.
- Add an IV surface (strike x expiry) heatmap.
- Historical Greeks tracking (save daily snapshots, chart Theta decay over the option's life).
- Connect to this project's macro/FX CIRP dashboard for a unified risk view.